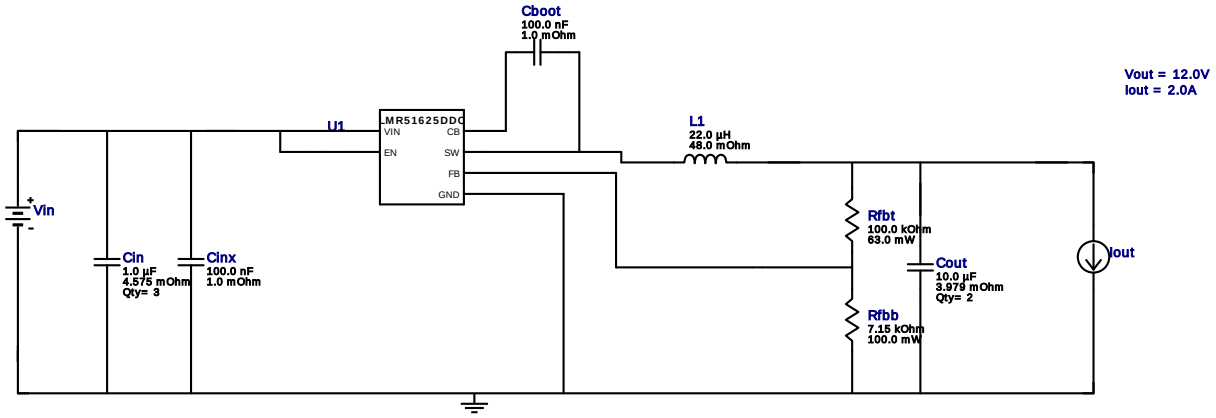
WEBENCH® Design Report

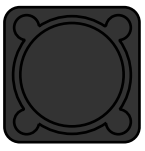
VinMin = 20.0V
VinMax = 52.0V
Vout = 12.0V
Iout = 2.0A

Device = LMR51625DDC
Topology = Buck
Created = 2026-04-05 19:36:11.399
BOM Cost = $2.02
BOM Count = 11
Total Pd = 1.23W

Design : 206 LMR51625DDC
LMR51625DDC 20V-52V to 12.00V @ 2A

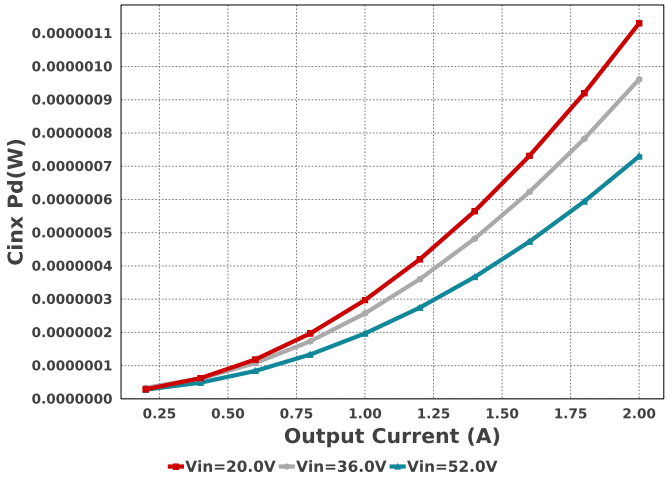


Electrical BOM

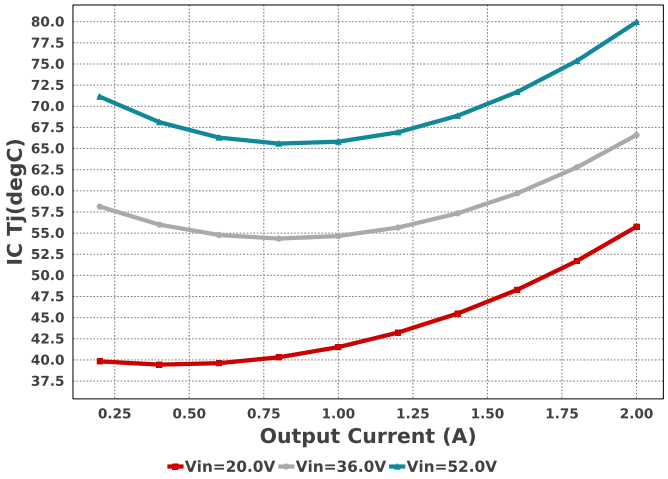
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	MuRata	GRM155R71C104KA88D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Cin	TDK	CGA5L2X7R2A105K160AA Series= X7R	Cap= 1.0 uF ESR= 4.575 mOhm VDC= 100.0 V IRMS= 3.39639 A	3	$0.14	1206_180 11 mm²
Cinx	MuRata	GRM188R72A104KA35D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 100.0 V IRMS= 3.85 A	1	$0.04	0603 5 mm²
Cout	Taiyo Yuden	MSASU32MSB5106KPNA01 Series= X5R	Cap= 10.0 uF ESR= 3.979 mOhm VDC= 50.0 V IRMS= 3.4821 A	2	$0.16	1210 15 mm²
L1	Bourns	SRR1208-220ML	L= 22.0 µH 48.0 mOhm	1	$0.56	 SRR1208 216 mm²
Rfbb	Yageo	RC0603FR-077K15L Series= ?	Res= 7.15 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²
Rfbt	Vishay-Dale	CRCW0402100KFKED Series= CRCW..e3	Res= 100.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
U1	Texas Instruments	LMR51625DDC	Switcher	1	$0.65	 DDC0006A 15 mm²

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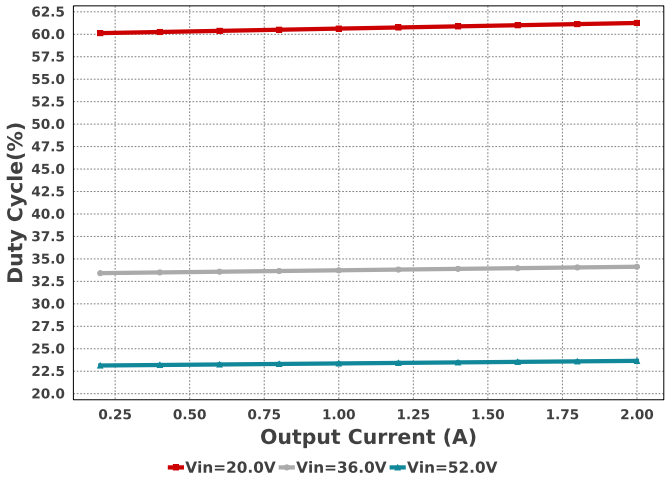
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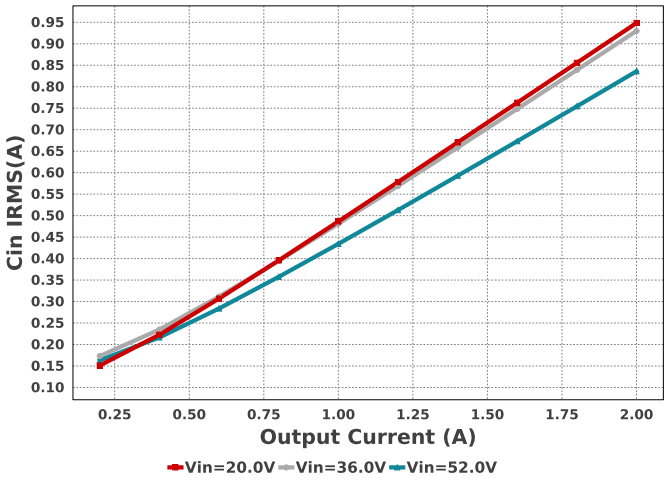
IC Tj



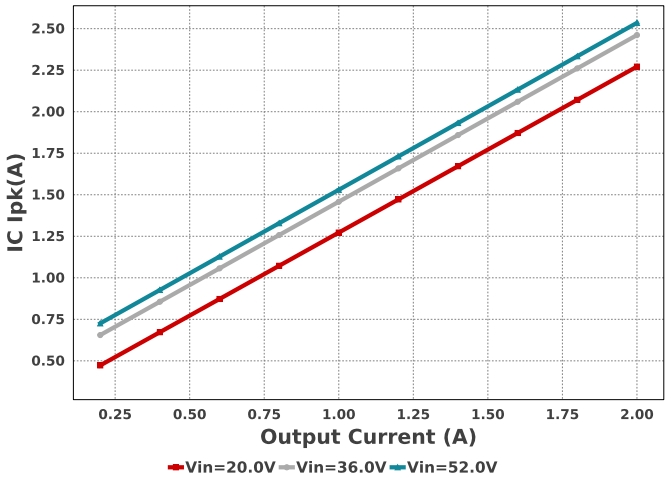
Duty Cycle



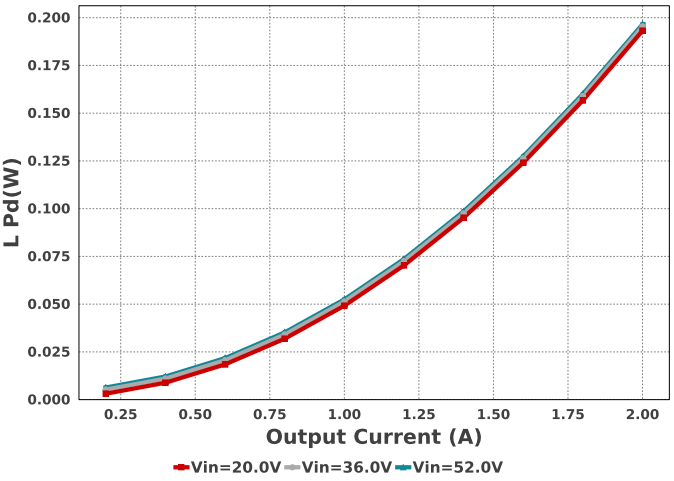
Cin IRMS



IC Ipk

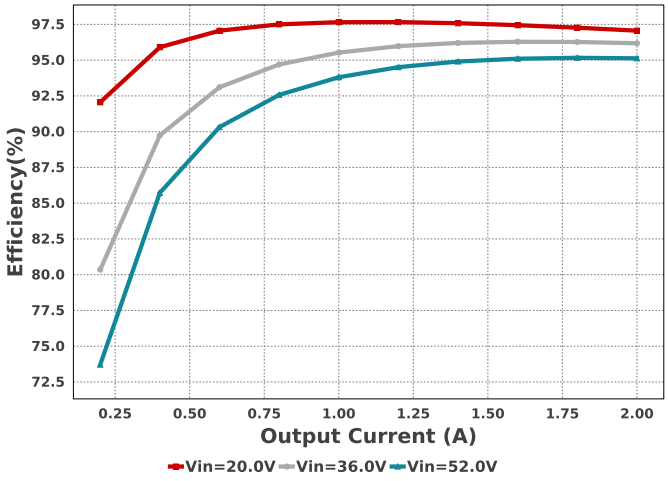


L Pd

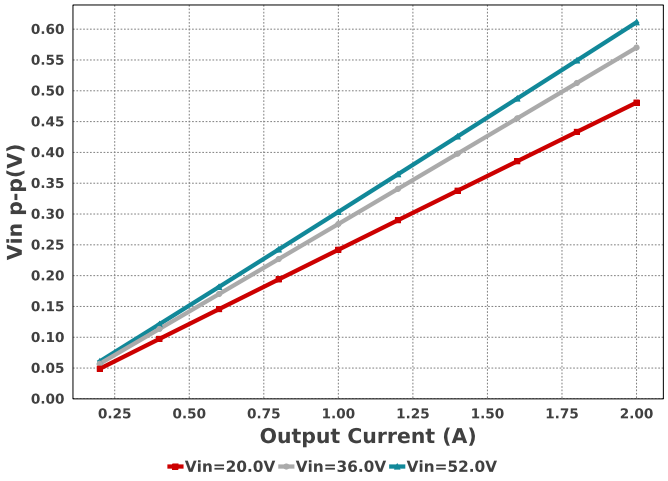


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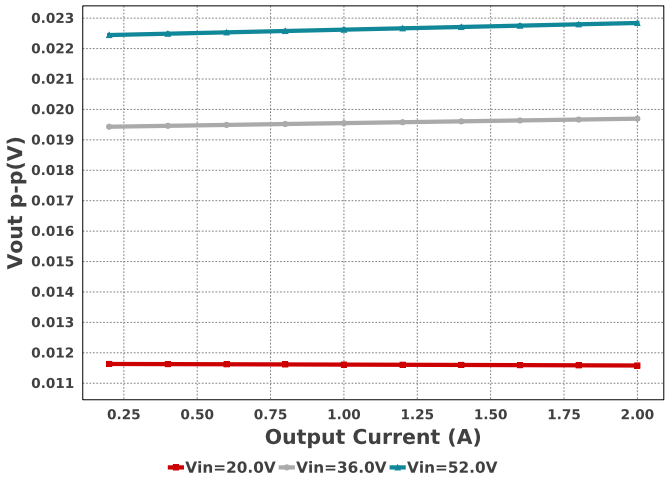
Efficiency



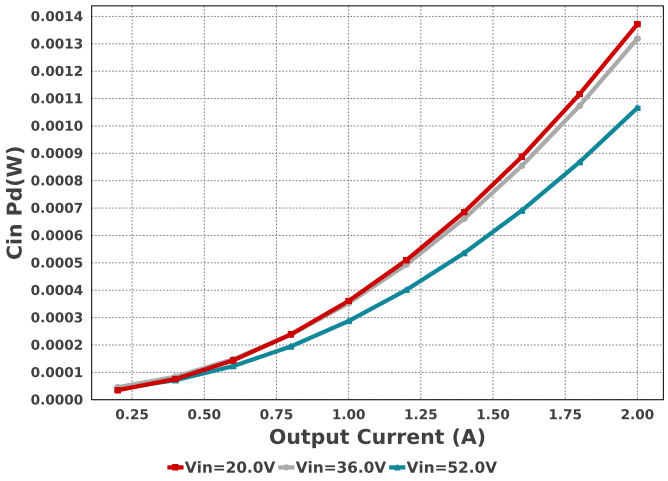
Vin p-p



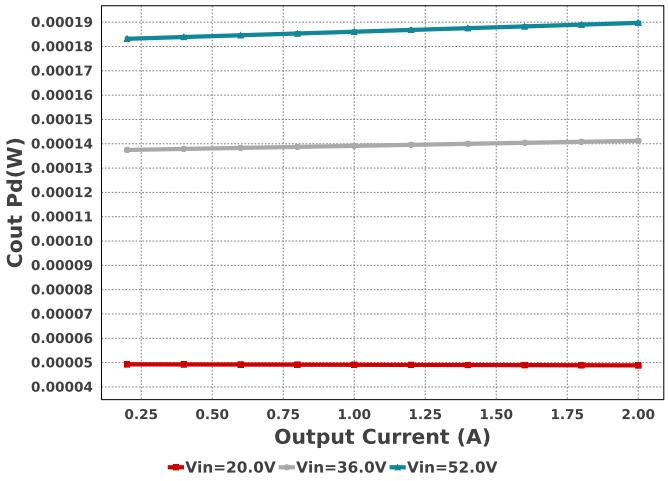
Vout p-p



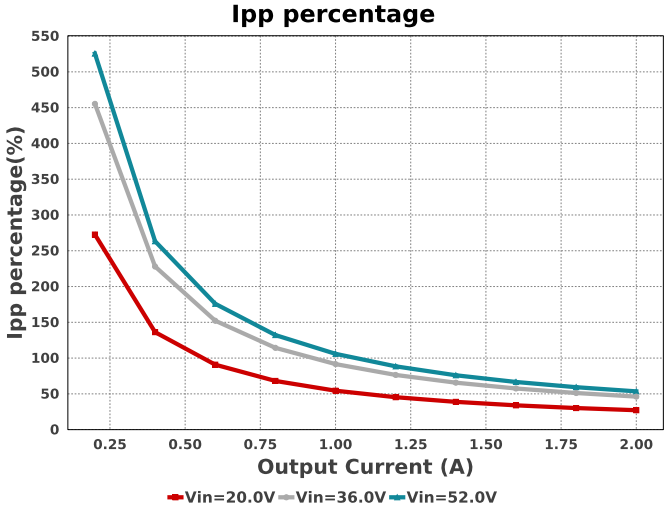
Cin Pd



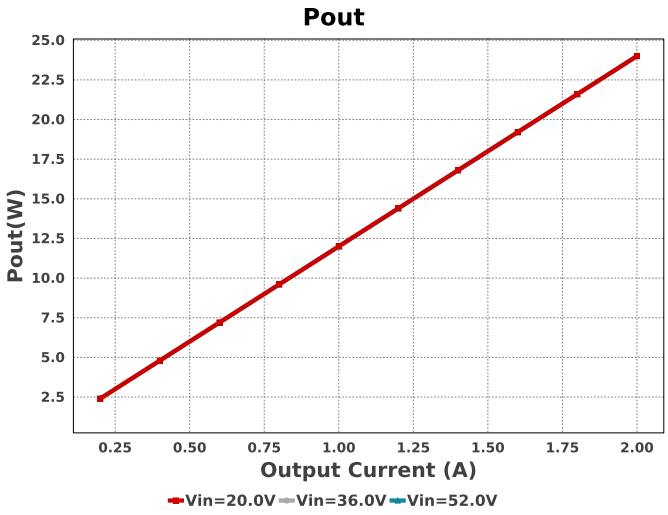
Cout Pd

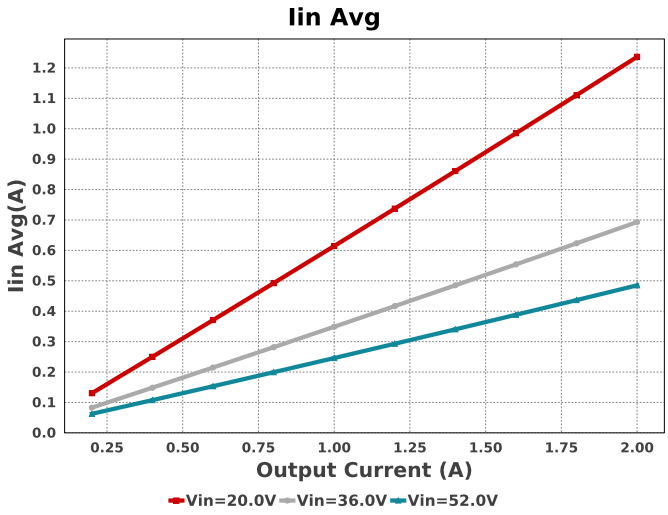


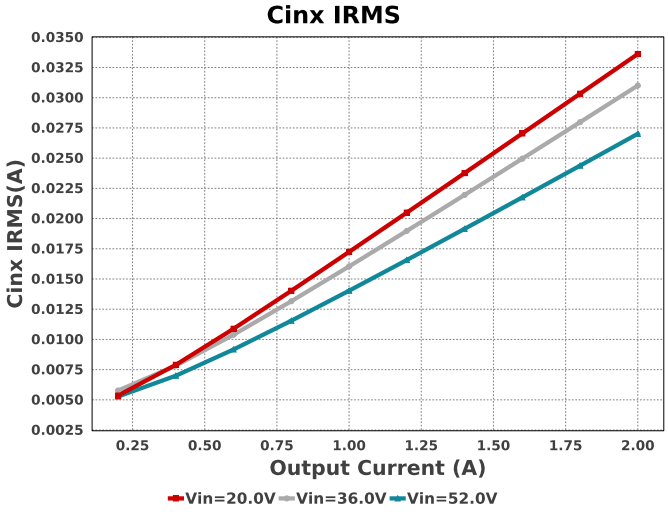
Ipp percentage

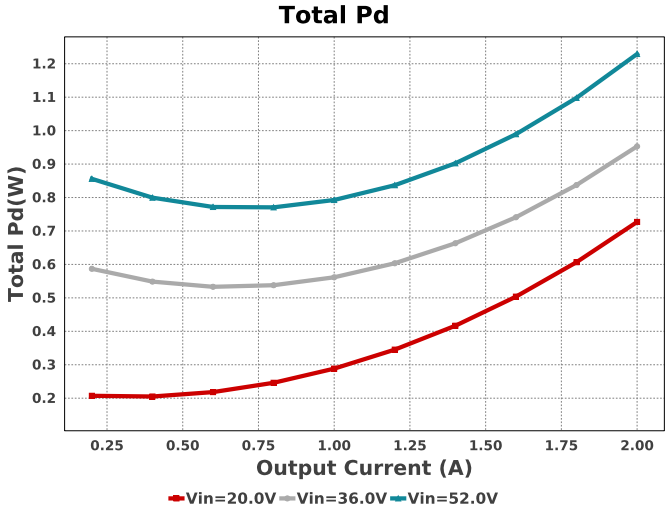


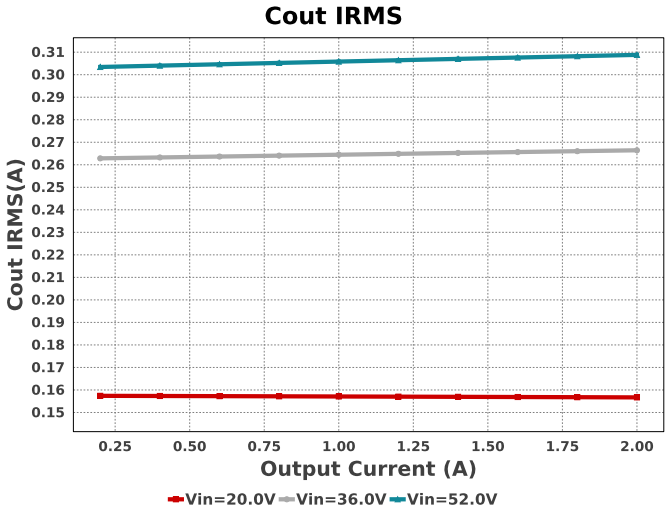
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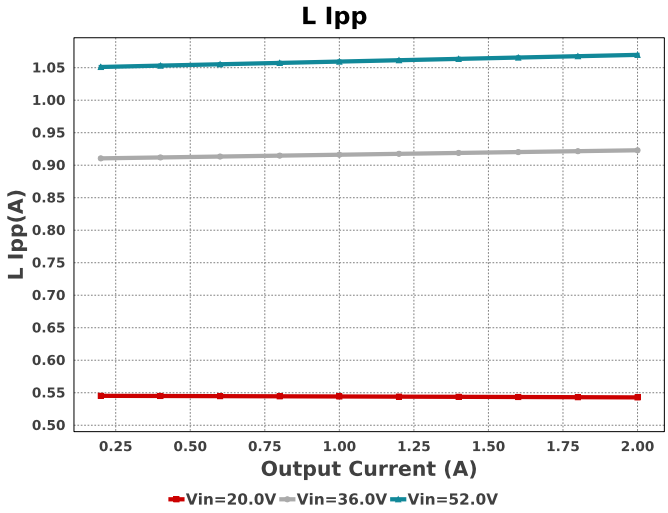




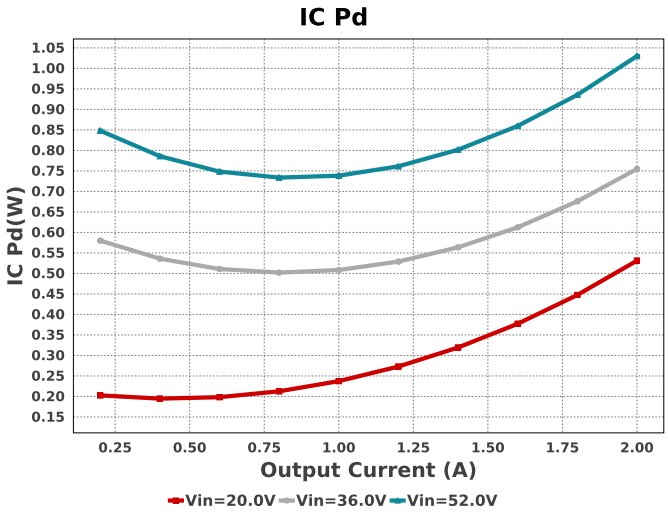


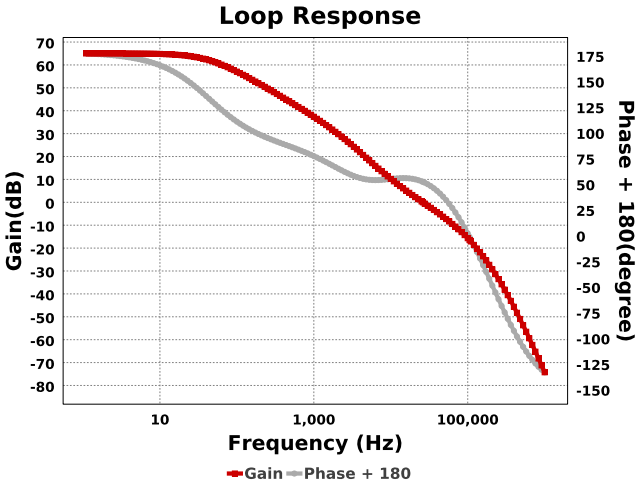






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Operating Values

#	Name	Value	Category	Description
1.	BOM Count	11		Total Design BOM count
2.	Total BOM	$2.023		Total BOM Cost
3.	Cin IRMS	836.062 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	1.066 mW	Capacitor	Input capacitor power dissipation
5.	Cinx IRMS	27.014 mA	Capacitor	Bulk capacitor RMS ripple current
6.	Cinx Pd	729.76 nW	Capacitor	Bulk capacitor power dissipation
7.	Cout IRMS	308.806 mA	Capacitor	Output capacitor RMS ripple current
8.	Cout Pd	189.72 µW	Capacitor	Output capacitor power dissipation
9.	IC Ipk	2.535 A	IC	Peak switch current in IC
10.	IC Pd	1.03 W	IC	IC power dissipation
11.	IC Tj	79.955 degC	IC	IC junction temperature
12.	IC Tolerance	80.0 mV	IC	IC Feedback Tolerance
13.	ICThetaJA Effective	48.5 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
14.	Iin Avg	485.18 mA	IC	Average input current
15.	Ipp percentage	53.487 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
16.	L Ipp	1.07 A	Inductor	Peak-to-peak inductor ripple current
17.	L Pd	196.58 mW	Inductor	Inductor power dissipation
18.	Cin Pd	1.066 mW	Power	Input capacitor power dissipation
19.	Cinx Pd	729.76 nW	Power	Bulk capacitor power dissipation
20.	Cout Pd	189.72 µW	Power	Output capacitor power dissipation
21.	IC Pd	1.03 W	Power	IC power dissipation
22.	L Pd	196.58 mW	Power	Inductor power dissipation
23.	Total Pd	1.229 W	Power	Total Power Dissipation
24.	Cross Freq	26.129 kHz	System Information	Bode plot crossover frequency
25.	Duty Cycle	23.653 %	System Information	Duty cycle
26.	Efficiency	95.128 %	System Information	Steady state efficiency
27.	FootPrint	308.0 mm^2	System Information	Total Foot Print Area of BOM components
28.	Frequency	400.0 kHz	System Information	Switching frequency
29.	Gain Marg	-16.502 dB	System Information	Bode Plot Gain Margin
30.	Iout	2.0 A	System Information	Iout operating point
31.	Iout transient step used for Cout calculations	1.5 A	System Information	Custom Transient current step requirement that was used for Cout selection (A).
32.	Low Freq Gain	65.023 dB	System Information	Gain at 1Hz
33.	Mode	CCM	System Information	Conduction Mode
34.	Overshoot Value	140.5 mV	System Information	Theoretical Vout Overshoot Value
35.	Phase Marg	53.547 deg	System Information	Bode Plot Phase Margin
36.	Pout	24.0 W	System Information	Total output power
37.	Undershoot Value	238.654 mV	System Information	Theoretical Vout Undershoot Value

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#	Name	Value	Category	Description
38.	Vin	52.0 V	System Information	Vin operating point
39.	Vin p-p	611.153 mV	System Information	Peak-to-peak input voltage
40.	Vout	12.0 V	System Information	Operational Output Voltage
41.	Vout Actual	11.989 V	System Information	Vout Actual calculated based on selected voltage divider resistors
42.	Vout Ripple requirement used for Cout calculations	3.0 %	System Information	Custom maximum output ripple requirement that was used for Cout selection(% of Vout).
43.	Vout Tolerance	12.074 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
44.	Vout p-p	22.841 mV	System Information	Peak-to-peak output ripple voltage
45.	Vout transient requirement used for Cout calculations	3.0 %	System Information	Custom Transient voltage change requirement that was used for Cout selection (% of Vout).

Design Inputs

Name	Value	Description
Iout	2.0	Maximum Output Current
VinMax	52.0	Maximum input voltage
VinMin	20.0	Minimum input voltage
Vout	12.0	Output Voltage
base_pn	LMR51625	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

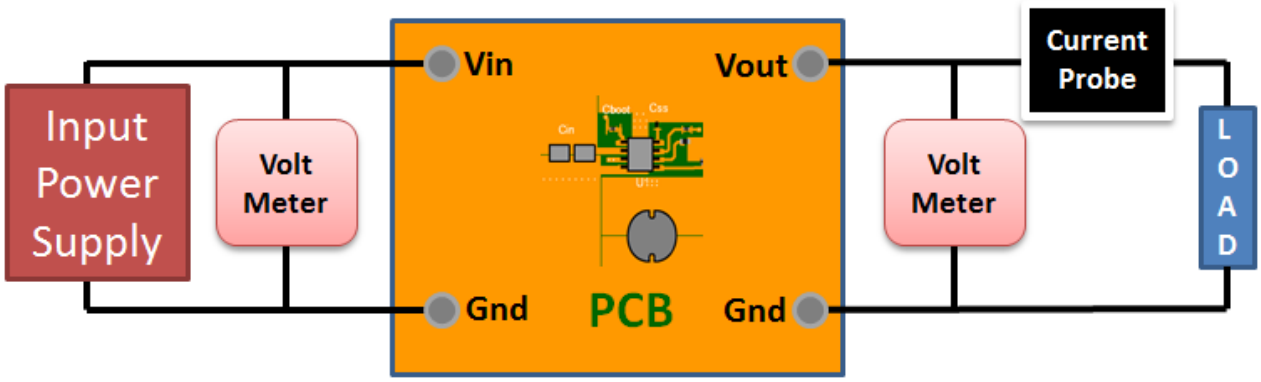
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 20.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 0901A20B76EF0432A028C0A579EB6467[v1]
2. **LMR51625** Product Folder : https://www.ti.com/product/LMR51625 : contains the data sheet and other resources.

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